

Learning theories: Piaget, Vygotsky, Bruner, Skinner

अधिगम सिद्धांत: पियाजे, वाइगोत्स्की, ब्रूनर, स्किनर

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Why this topic matters for the Vice Principal test

These four names are the core of B.Ed. learning theory, and objective questions test them by swapping their ideas across each other. The 2026 CRT syllabus (Advt 51/2026) lists "Pedagogy in Education" as one of its eight topics. The closest earlier paper is the 2022 Vice Principal recruitment test (Advt 10/2022). Its file name suggests a General Ability plus Education format, and the education part of such papers leans on B.Ed. psychology: stages, key terms, famous quotations and classroom applications. The questions are rarely deep. The traps are almost always attribution (who said what), sequence (which stage comes first) and sign (reinforcement versus punishment).

Read this pack in order. The three schools come first, then each theorist, then a comparison, then the traps.

The three schools at a glance

**Behaviourism studies observable behaviour,
cognitivism studies mental processes,**

constructivism studies how learners build knowledge.

School	Core claim	Key names
Behaviourism	Learning = change in observable behaviour through stimulus, response and consequence	Pavlov, Thorndike, Watson, Skinner
Cognitivism	Learning = change in mental structures; insight, memory, information processing	Köhler, Tolman, Bruner , Ausubel, Gagné
Constructivism	Learner actively constructs knowledge from experience	Piaget (cognitive), Vygotsky (social)

Bruner sits on the border of cognitivism and constructivism. Exam setters usually label him a **cognitive psychologist** and a founder of **discovery learning**.

Jean Piaget (1896–1980): cognitive constructivism

Piaget, a Swiss biologist turned psychologist, argued that children build knowledge in a fixed sequence of four stages and that development comes before learning. He called his field **genetic epistemology** (the study of how knowledge grows). He used the **clinical method**, observing and questioning children, including his own three. He headed the International Bureau of Education in Geneva for decades, which is why his ideas reached school systems so quickly.

Core concepts

Term	Meaning	Classroom example
Schema	Mental structure or pattern for organising knowledge	A child's "dog" schema: four legs, fur, barks
Assimilation	Fitting new information into an existing schema	Calls a cat "dog"
Accommodation	Changing a schema to fit new information	Learns a new category "cat"
Equilibration	Balancing assimilation and accommodation; disequilibrium drives learning	Confusion about a floating stone pushes the child to revise ideas

Adaptation is the umbrella term for assimilation plus accommodation. **Organisation** is the tendency to link schemas into systems.

The four stages

The **sequence is invariant and universal**, but ages are approximate. A child cannot skip a stage.

Stage and age	Hallmarks	Limits
Sensorimotor (0–2)	Learns through senses and movement; object permanence emerges around 8 months; deferred imitation and first symbols near age 2	No mental representation at first
Preoperational (2–7)	Symbolic function: language, pretend play; intuitive thought	Egocentrism, animism, centration, irreversibility, no conservation
Concrete operational (7–11)	Conservation, reversibility, decentration, seriation, classification, transitivity	Logic only on concrete objects, not on hypotheses
Formal operational (11 onward)	Abstract, hypothetico-deductive, propositional reasoning; systematic testing of variables	Adolescent egocentrism (imaginary audience, personal fable, described by Elkind)

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`mermaid flowchart TD
  A["Sensorimotor (0–2):  
object permanence"] --> B["Preoperational (2–7):  
egocentrism"]
  B --> C["Concrete operational (7–11):  
conservation"]
  C --> D["Formal operational (11+):  
abstract reasoning"]
  `

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Vocabulary UPSC expects you to decode:

- **Object permanence:** knowing an object exists when out of sight (peek-a-boo works because the baby lacks it).
- **Egocentrism:** inability to take another person's viewpoint; tested with the **three mountains task** (Piaget and Inhelder).
- **Animism:** giving life and feelings to objects ("the sun is tired").
- **Centration:** focusing on one feature only, such as the height of a glass and not its width.
- **Conservation:** understanding that quantity stays the same when shape or arrangement changes. Number is conserved first (about age 6–7), then mass and liquid, weight (about 9) and volume (about 11–12). This staggered mastery within one stage is called **horizontal décalage**.
- **Reversibility:** mentally undoing an action (if $3 + 4 = 7$, then $7 - 4 = 3$).
- **Seriation:** ordering by size or another dimension.
Transitivity: if $A > B$ and $B > C$ then $A > C$.
- **Hypothetico-deductive reasoning:** forming hypotheses and testing them systematically, as in the **pendulum task**.

Piaget on moral development

Piaget's *The Moral Judgment of the Child* (1932)

describes a shift from **heteronomous morality** (moral realism: rules are fixed, judged by consequences, belief in immanent justice) to **autonomous morality** (moral relativism: rules are agreements, intentions

matter). **Kohlberg** later built his six stages on this base.

Educational implications and criticisms

Implications: match teaching to the child's stage (**readiness**); use **concrete materials** before abstractions; favour **active discovery**; use **peer interaction** to create disequilibrium; accept that errors reveal thinking. The teacher is a facilitator, not a transmitter.

Criticisms: he **underestimated** young children (later work shows object permanence and some perspective taking appear earlier); stages are less discrete than claimed; many adults never show consistent formal operations; he ignored **culture and instruction**; his samples were small. These criticisms are exactly what Vygotsky's theory answers.

Lev Vygotsky (1896–1934): sociocultural theory

Vygotsky, a Russian (Soviet) psychologist who died of tuberculosis at 37, held that learning is social first and individual second, so good teaching can pull development forward. His major work *Thought and Language* appeared in 1934, the year he died, and reached the West in English only in 1962. *Mind in Society* (1978) collected his essays. Because his work was suppressed in the Soviet Union for years, his influence peaked long after his death.

Core concepts

- **Social constructivism / sociocultural theory:**
knowledge is constructed through interaction with

others within a culture.

- **General genetic law of cultural development:** every higher function appears **twice**, first between people (**interpsychological**) and then inside the child (**intrapsychological**). This movement inward is **internalisation**.
- **Cultural tools and psychological tools:** language, number systems, writing, maps and signs **mediate** thinking. Language is the most important tool.
- **Elementary versus higher mental functions:** elementary functions are biological (basic perception, involuntary attention); higher functions (voluntary attention, logical memory, concept formation) are culturally built through tools.
- **Zone of Proximal Development (ZPD):** the gap between what a learner can do **alone** and what the learner can do **with guidance** from a **More Knowledgeable Other (MKO)**, who may be an adult or a more capable peer. Teaching should aim inside the ZPD, not at what the child already does or at what is far beyond reach.
- **Scaffolding:** temporary support that is gradually withdrawn as competence grows. Note the attribution trap: the word was introduced by **Wood, Bruner and Ross (1976)**, not by Vygotsky, though it describes his idea.
- **Learning leads development:** the opposite of Piaget. "What a child can do with assistance today she will be able to do by herself tomorrow."

- **Private speech:** children talking to themselves while working. Vygotsky saw it as a **tool of thought and self-regulation** that later becomes **inner speech**. Piaget called the same behaviour **egocentric speech** and treated it as immaturity that fades. The Vygotskian sequence is **social speech → private (egocentric) speech → inner speech**.

Classroom applications

Cooperative learning, peer tutoring, reciprocal teaching, guided participation, dynamic assessment (assessing what the child can do with help, not only alone), and **differentiated instruction** that targets each learner's ZPD. The teacher is a mediator and co-creator. Multilingual, culturally rooted teaching, including use of the home language in the early grades, fits Vygotsky's view of language as the main tool of thought.

Piaget versus Vygotsky

Both are constructivists; Piaget is individual and biological, Vygotsky is social and cultural.

Dimension	Piaget	Vygotsky
Source of knowledge	Child's own action on objects	Social interaction, culture
Learning and development	Development precedes learning	Learning leads development
Stages	Four universal stages	No fixed stages; ZPD varies
Language	Product of thought	Tool that shapes thought
Child's self-talk	Egocentric, fades	Private speech, becomes inner speech
Role of teacher	Facilitator; provide materials and readiness	Mediator; scaffold within ZPD
Peers	Create cognitive conflict	More capable peers teach

Jerome Bruner (1915–2016): discovery, structure and the spiral

Bruner, an American cognitive psychologist, is the name behind discovery learning, the spiral curriculum and the three modes of representation, and he co-coined the term scaffolding. He co-founded the **Harvard Center for Cognitive Studies (1960)** with George Miller and helped start the "cognitive revolution" against behaviourism. He also wrote the introduction to the 1962 English edition of Vygotsky's *Thought and Language*, carrying Vygotsky's ideas into Western classrooms.

Key works and ideas

Work	Year	Idea
<i>A Study of Thinking</i> (with Goodnow and Austin)	1956	Concept formation and concept attainment strategies; basis of the Concept Attainment Model
<i>The Process of Education</i>	1960	Report of the 1959 Woods Hole conference; structure of the discipline; readiness; intuitive and analytic thinking; motives for learning
<i>Toward a Theory of Instruction</i>	1966	Three modes of representation; four features of a theory of instruction
<i>Actual Minds, Possible Worlds</i>	1986	Narrative versus paradigmatic (logico-scientific) modes of thought
<i>The Culture of Education</i>	1996	Education as cultural meaning making; "folk pedagogy"

Three modes of representation: **enactive** (knowing through action, as in tying a knot), **iconic** (through images and diagrams), **symbolic** (through language, numbers and symbols). Unlike Piaget's stages, these modes **coexist** in adults and the teacher should move through them in that order when introducing a new concept.

Spiral curriculum: key ideas are revisited at increasing depth across grades, so the structure of a discipline is grasped early and refined later. His famous claim from *The Process of Education*: "**Any subject can be taught**

effectively in some intellectually honest form to any child at any stage of development."

Discovery learning: learners find principles themselves through guided inquiry rather than reception. Bruner expected it to raise intrinsic motivation, memory and problem-solving transfer. Ausubel's **meaningful reception learning** with **advance organisers** is the standard counter position.

Four features of a theory of instruction (1966): the learner's **predisposition** to learn, the **structure** of knowledge so that it can be grasped, the most effective **sequence** of presentation, and the nature and pacing of **rewards and punishments** (reinforcement).

Bruner's **MACOS** project (*Man: A Course of Study*) applied these ideas to a social studies curriculum. In later life he moved closer to Vygotsky and stressed culture, narrative and meaning.

B. F. Skinner (1904–1990): operant conditioning
Skinner, an American radical behaviourist, explained learning through consequences: behaviour followed by reinforcement is strengthened, behaviour followed by punishment or no consequence weakens. He built the **Skinner box** (operant chamber) for rats and pigeons and the first widely known **teaching machine**. Key works: *The Behavior of Organisms* (1938), *Walden Two* (1948), *Science and Human Behavior* (1953), the article "The Science of Learning and the Art of Teaching" (1954), *Verbal*

Behavior (1957, attacked by Chomsky in 1959), *The Technology of Teaching* (1968) and *Beyond Freedom and Dignity* (1971).

Respondent versus operant

Pavlov's classical conditioning is **respondent** or **Type S** conditioning: a stimulus elicits an involuntary response. Skinner's operant conditioning is **Type R** or **instrumental**: the organism emits a voluntary response and the **consequence** determines whether it recurs. Thorndike's **Law of Effect** was the precursor.

Reinforcement and punishment: the 2 × 2 that UPSC loves

	Stimulus added	Stimulus removed
Behaviour increases	Positive reinforcement (praise, stars, marks)	Negative reinforcement (homework completed, so no detention; seat belt stops the buzzer)
Behaviour decreases	Positive punishment (scolding, extra work)	Negative punishment (time-out, loss of play period, response cost)

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`mermaid flowchart TD
  A["Consequence follows behaviour"] --> B["Stimulus added (positive)"]
  A --> C["Stimulus removed (negative)"]
  B --> D["Behaviour increases: positive reinforcement"]
  C --> E["Behaviour decreases: positive punishment"]
  B --> F["Behaviour increases: negative reinforcement"]
  C --> G["Behaviour decreases: negative punishment"]
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Rules to remember: **positive** means something is added, **negative** means something is taken away.

Reinforcement always increases behaviour, **punishment always decreases** it. Negative reinforcement is **not** punishment. Skinner himself argued that punishment is weak and temporary and preferred **extinction** plus positive reinforcement of alternatives.

Other terms: **primary reinforcers** (food, water) versus **secondary or conditioned reinforcers** (marks, money, tokens); **shaping** by rewarding **successive approximations** of the target behaviour; **chaining** linking behaviours in sequence; **extinction** when reinforcement stops; **spontaneous recovery**; **stimulus generalisation** and **discrimination**. **Token economy** and **behaviour modification** are classroom applications. The **Premack principle** (do the less preferred activity to earn the preferred one, "grandma's rule") is from David Premack, not Skinner.

Schedules of reinforcement

Schedule	Reinforce after...	Pattern
Continuous	Every correct response	Fast learning, fast extinction
Fixed ratio (FR)	Set number of responses (every 5th problem)	High rate, pause after reward
Variable ratio (VR)	Unpredictable number (gambling)	Highest, steadiest rate; most resistant to extinction
Fixed interval (FI)	Set time (weekly test)	Scallop: slow then rushed before deadline
Variable interval (VI)	Unpredictable time (surprise quiz)	Moderate, steady rate

The **partial reinforcement effect**: behaviour learnt under intermittent schedules resists extinction more than behaviour learnt under continuous reinforcement. Ratio schedules give higher response rates than interval schedules.

Programmed instruction and teaching machines

Skinner's **linear programmed instruction** rests on **small steps, active responding, immediate feedback (confirmation), self-pacing** and a **low error rate**.

Norman Crowder devised **branching (intrinsic) programming** with multiple-choice frames and remedial loops. **Sidney Pressey** had built a mechanical testing machine in the 1920s, but Skinner's 1954 paper made teaching machines famous. **Mastery learning**

and Keller's Personalised System of Instruction are descendants. So are drill-and-practice apps and adaptive learning software today.

Classroom applications and limits

Behaviourist techniques: clear behavioural objectives (Mager), **contingency contracts**, token economies, time-out, response cost, praise schedules, immediate feedback, drill for fluency. Limits: ignores meaning, motivation and understanding; extrinsic rewards can undermine intrinsic motivation; Chomsky showed that language cannot be explained by reinforcement alone.

Other names that appear beside these four

UPSC often adds a fifth option from this list, so learn the one-line tag for each.

Theorist	Tag
Pavlov	Classical conditioning; dog, bell, salivation; UCS, CS, CR
Thorndike	Trial and error; connectionism; laws of readiness, exercise, effect ; cat in puzzle box
Watson	Founder of behaviourism; Little Albert
Köhler	Insight learning ; chimpanzee Sultan; Gestalt
Tolman	Latent learning, cognitive maps ; sign learning
Bandura	Social learning / observational learning; Bobo doll ; attention, retention, reproduction, motivation; self-efficacy
Ausubel	Meaningful reception learning; advance organisers
Gagné	Eight types of learning hierarchy; nine events of instruction
Kohlberg	Six stages of moral reasoning in three levels
Erikson	Eight psychosocial stages

Link to NEP 2020, NCF and the Delhi classroom

The 5+3+3+4 design of NEP 2020 is a stage theory in policy form. NEP 2020 (para 4.1) says the curricular and pedagogical structure must correspond to the developmental needs of learners at ages **3–8, 8–11, 11–14 and 14–18**. The Foundational stage is **play and activity based** (Piaget's preoperational child, Bruner's enactive mode). The Preparatory stage adds **discovery and interactive classroom learning** (concrete operations, iconic mode). The Middle stage introduces **experiential learning within subjects** and the

Secondary stage **critical thinking and multidisciplinary study** (formal operations, symbolic mode). The mapping to Piaget and Bruner is an interpretation, not wording from NEP.

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`mermaid flowchart TD
  A["Foundational (3–8): play based"] --> B["Preparatory (8–11): discovery learning"]
  B --> C["Middle (11–14): experiential learning"]
  C --> D["Secondary (14–18): critical thinking"]`
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NCF 2005 made **constructivism** the official pedagogy of Indian schools and criticised rote learning; the **NCF for the Foundational Stage (2022)** and the **NCF for School Education (2023)** keep this, with explicit use of play, toys, stories and the home language, which fits Vygotsky's social and language-based view of learning. **NIPUN Bharat (2021)**, the mission for foundational literacy and numeracy, likewise stresses play-based and activity-based learning. In Delhi government schools the earlier activity-based programmes, the **Happiness Curriculum** and **Mission Buniyaad** (both 2018), were replaced in **September 2025** by new programmes, including **Science of Living**, **NEEEV** and **Rashtraneeti**. A Vice Principal supervises this kind of experiential teaching. Competency-based assessment under **PARAKH** asks for application, not recall, which is a Bruner and Piaget position rather than a Skinner one.

Common traps

- **Scaffolding is not Vygotsky's word.** Wood, Bruner and Ross coined it in 1976. ZPD is Vygotsky's;

scaffolding is the teaching strategy built on ZPD.

- **Negative reinforcement is not punishment.** Negative reinforcement removes something unpleasant and increases behaviour. Time-out and response cost are negative punishment.
- **Development precedes learning (Piaget) versus learning leads development (Vygotsky).** Option writers reverse them.
- **Bruner's modes are not stages.** Enactive, iconic and symbolic coexist; Piaget's four stages are sequential and age linked.
- **Spiral curriculum is Bruner,** not Piaget, even though it uses readiness.
- **Conservation belongs to concrete operations,** not to the preoperational stage; the preoperational child fails conservation.
- **Object permanence is sensorimotor,** not preoperational.
- **Hypothetico-deductive reasoning is formal operational;** transitivity and seriation are concrete operational.
- **Egocentric speech:** immaturity for Piaget, tool of thought for Vygotsky.
- **Type S is Pavlov, Type R is Skinner.** Thorndike's Law of Effect is the ancestor of operant conditioning. The word "reinforcement" itself already appears in Pavlov's classical conditioning; Skinner made it the centre of operant theory.
- **Variable ratio** gives the highest rate and most resistance to extinction; **fixed interval** gives the

scallop.

- **Premack principle** is Premack, **Little Albert** is Watson, **Bobo doll** is Bandura, **insight** is Köhler. None of these is Skinner.
- **Branching programmed instruction** is Crowder, linear is Skinner.
- The quotation "**any subject can be taught... to any child at any stage**" is Bruner (1960), often misattributed to Piaget or Dewey.
- **Piaget's moral stages (heteronomous, autonomous)** are two; Kohlberg's are six.

Remember for the exam

1. Piaget: Swiss, 1896–1980, genetic epistemology, four stages, development before learning.
2. Stage ages: sensorimotor 0–2, preoperational 2–7, concrete operational 7–11, formal operational 11 onward.
3. Assimilation fits new data into old schemas; accommodation changes the schema; equilibration balances the two.
4. Object permanence (about 8 months) is sensorimotor; conservation, reversibility, seriation, transitivity are concrete operational.
5. Egocentrism is tested by the three mountains task; formal operations by the pendulum task.
6. Vygotsky: Russian (Soviet), 1896–1934, *Thought and Language* 1934, sociocultural theory, learning leads development.

7. ZPD = gap between unaided and assisted performance; MKO guides; scaffolding (Wood, Bruner, Ross 1976) is the support.
8. Speech sequence for Vygotsky: social → private (egocentric) → inner.
9. Bruner: American, 1915–2016, *The Process of Education* 1960, spiral curriculum, discovery learning.
10. Bruner's modes: enactive, iconic, symbolic, in that teaching order, and they coexist.
11. Bruner's theory of instruction (1966): predisposition, structure, sequence, reinforcement.
12. Skinner: American, 1904–1990, operant (Type R) conditioning, Skinner box, *The Behavior of Organisms* 1938.
13. Positive = add, negative = remove; reinforcement increases, punishment decreases.
14. Variable ratio: highest and steadiest rate, most resistant to extinction; fixed interval: scallop.
15. Programmed instruction: Skinner linear, Crowder branching; Pressey built an earlier testing machine in the 1920s.
16. NEP 2020 stages 3–8, 8–11, 11–14, 14–18 map onto Piaget's stages and Bruner's modes; NCF 2005 made constructivism official.

Quick facts

- Piaget (Swiss, 1896–1980) called his field genetic epistemology and held that development

precedes learning.

- Piaget's four stages: sensorimotor 0–2, preoperational 2–7, concrete operational 7–11, formal operational 11 onward; the sequence is invariant, ages are approximate.
- Assimilation fits new information into an existing schema; accommodation changes the schema; equilibration balances the two.
- Object permanence (around 8 months) is sensorimotor; conservation, reversibility, seriation and transitivity are concrete operational; hypothetico-deductive reasoning is formal operational.
- Egocentrism is tested by the three mountains task; a preoperational child shows animism, centration and irreversibility.
- Vygotsky (Russian/Soviet, 1896–1934) wrote *Thought and Language* (1934); his sociocultural theory says learning leads development.
- Zone of Proximal Development = gap between what a learner does alone and with help from a More Knowledgeable Other.
- The term scaffolding was coined by Wood, Bruner and Ross in 1976, not by Vygotsky.
- Vygotsky's speech sequence: social speech → private (egocentric) speech → inner speech; Piaget saw egocentric speech as immaturity.
- Bruner (American, 1915–2016): discovery learning, spiral curriculum, and three modes of

representation (enactive, iconic, symbolic) that coexist rather than replace each other.

- Bruner's *The Process of Education* (1960): 'Any subject can be taught effectively in some intellectually honest form to any child at any stage of development.'
- Skinner (American, 1904–1990): operant (Type R) conditioning, Skinner box, *The Behavior of Organisms* (1938), teaching machine paper 1954.
- Positive means a stimulus is added and negative means removed; reinforcement always increases behaviour, punishment always decreases it; negative reinforcement is not punishment.
- Variable ratio schedule gives the highest, steadiest response rate and most resistance to extinction; fixed interval gives the scallop pattern.
- Programmed instruction: Skinner's linear programme (small steps, active response, immediate feedback) versus Crowder's branching programme.